

AICTE | IBM SkillsBuild Internship Project Report

Project Title: Supermarket Sales Analysis and Predictive AI Modeling

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1. Abstract

In modern retail operations, data analytics and machine learning enable demand forecasting and profitability optimization. This project evaluates customer purchasing patterns across supermarket branches and demonstrates predictive modeling on transaction revenues using exploratory data analysis and supervised learning.

2. Objectives

- Clean and preprocess multi-branch sales records.
- Perform Exploratory Data Analysis (EDA) on product performance, customer tiers, and branch revenue.
- Train a machine learning regression model to predict sales totals accurately.

3. Methodology

- **Data Cleaning & Encoding:** Categorical encoding applied across branches, product categories, and customer membership tiers using Scikit-Learn's `LabelEncoder`.
- **Exploratory Data Analysis (EDA):** Computed summary statistics and correlation trends showing the relationship between transaction price, quantity, and customer ratings.
- **AI / ML Model:** Implemented a Random Forest Regressor with standard scaling (`StandardScaler`) on an 80/20 train-test split.

4. Results & Business Recommendations

- **Model Accuracy:** Achieved an R^2 score of 0.9996 with an RMSE of ~4.55, demonstrating high reliability in estimating purchase amounts.
- **Recommendations:** Implement targeted loyalty programs for non-member customers to convert them into member status, and optimize inventory for high-velocity product categories.